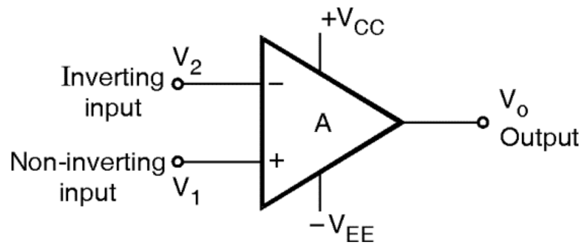
Basic Information of Op-Amp

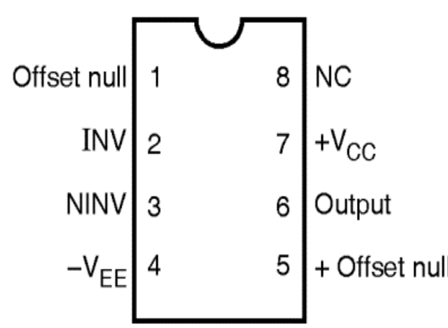
- The Operational Amplifier is a multi terminal device which has advantages like small size with reduced cost and power consumption and easy to replace.
- Op-amp is a versatile device that can be used to amplify both ac and dc input signals.
- Along with amplification it performs various mathematical operations such as addition, subtraction, multiplication, differentiation, integration and the list is on and on due to which, it is named as OPERATIONAL AMPLIFIER.
- Figure shows symbol for an op-amp along with its various terminals (two input and one output terminals)



Symbol of Op-amp

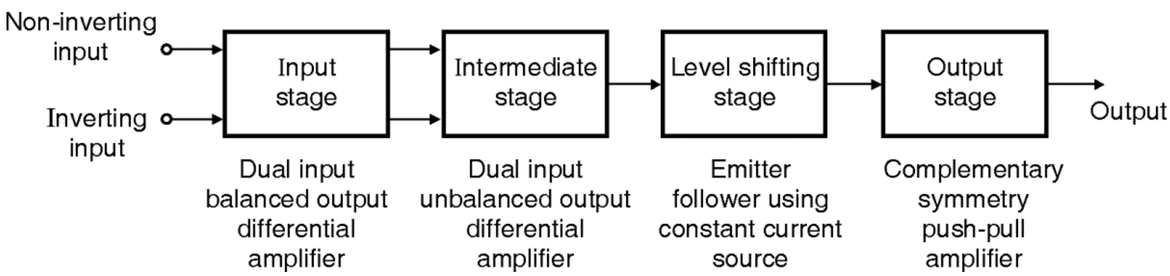
- As shown in figure Op-amp consists of two input terminals namely Inverting input and Non-inverting input with a single output terminal

Pin Diagram of Op-amp IC 741



Block Diagram of Op-Amp

- An operational amplifier is a direct coupled high gain multistage amplifier usually consisting of one or more differential amplifiers followed by a level translator and output stage.
- It can be represented by a block diagram as shown in figure



Block Diagram of Op-Amp

As shown in figure 1.4, op-amp consists of Three stages as

1. Input Stage.
2. Intermediate Stage.
    - Intermediate Stage
    - Level Shifting Stage
3. Output Stage

- The heart of op-amp is a differential amplifier which amplifies difference between input applied to non-inverting and inverting terminals.

INPUT STAGE :

- The input stage is the dual-input, balanced output differential amplifier.
- Here both inverting and noninverting input is given to input stage.
- It provides high voltage gain of amplifier.
- Input stage has high input resistance of op-amp.
- The balanced output of input stage is applied to intermediate stage

INTERMEDIATE STAGE :

- The intermediate stage is dual-input unbalanced output differential amplifier.
- Output of input stage is applied as input to intermediate stage.
- The unbalanced output of intermediate stage is applied to level shifting stage

LEVEL SHIFTING STAGE :

- As direct coupling is used in earlier stages, the dc voltage at the intermediate stage is well above ground potential.
- Due to this the level shifting stage is used after intermediate stage to shift the dc level downward to zero with respect to ground.
- Here emitter follower circuit also called as Buffer is used.

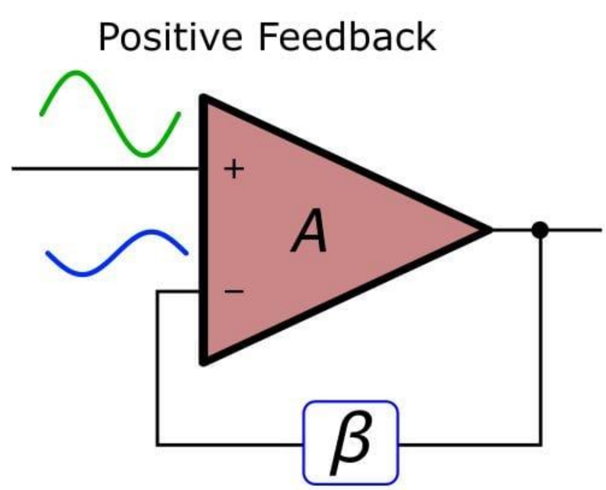
OUTPUT STAGE :

- The output stage increases output voltage and raises the current supplying capability of op-amp.
- It also provides low output resistance.
- Output stage is usually a push-pull complementary amplifier.

Concept of Feedback in Opamp

- In an Operational Amplifier (Op-Amp), feedback is the process of returning a portion of the output signal back to one of the input terminals.
- As Op-Amps have extremely high open-loop gain , they are rarely used without feedback except in comparator applications.

1. Positive Feedback



- Positive feedback occurs when the output is fed back to the non-inverting (+) input terminal.
- It is called "**regenerative**" feedback because the feedback signal reinforces the input signal.

Key Characteristics

- Instability: It drives the op-amp into saturation (the output "slams" to either the positive or negative power rail).
- Bistable Behavior: The output usually has only two stable states (High or Low)
- Hysteresis: It creates a "memory" effect where the switching threshold changes depending on the current output state. This prevents rapid switching (chatter) caused by noise.
- Regenerative Gain: It increases the gain to the point of instability.

Mathematical Expression

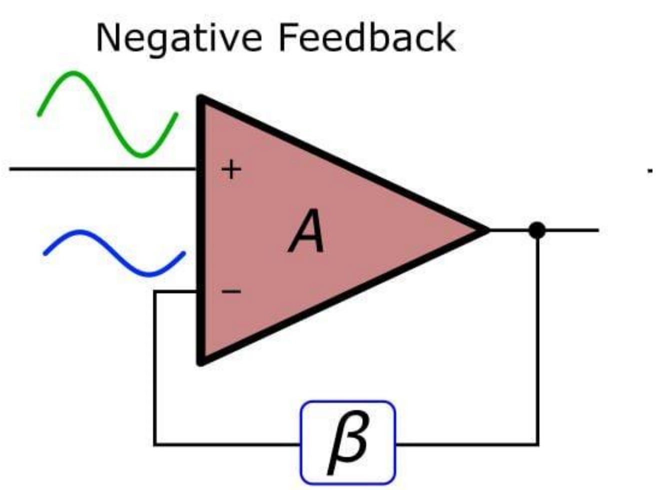
The gain with positive feedback is:

$$A_f = \frac{A}{1 - A\beta}$$

Applications

- Schmitt Trigger: A comparator with hysteresis used to clean up noisy signals.
- Oscillators: Circuits like the Wien Bridge or Relaxation Oscillator that generate waveforms (Sine, Square, etc.) without an external input.

2. Negative Feedback



- Negative feedback occurs when a portion of the output is fed back to the inverting (–) input terminal.
- It is called "**degenerative**" feedback because the feedback signal opposes the original input signal

Key Characteristics

- Stability: It "tames" the high gain of the op-amp, making the circuit stable and predictable.
- Virtual Short/Ground: In a negative feedback loop, the op-amp adjusts its output to keep the voltage difference between the two inputs near zero
- Gain Control: The gain is determined by external resistors rather than the op-amp's internal characteristics.
- Increased Bandwidth: Reducing the gain increases the frequency range over which the amplifier can operate.
- Reduced Distortion: It improves linearity and reduces noise.

Mathematical Expression

The gain with negative feedback is:

$$\frac{A}{1 + A\beta}$$

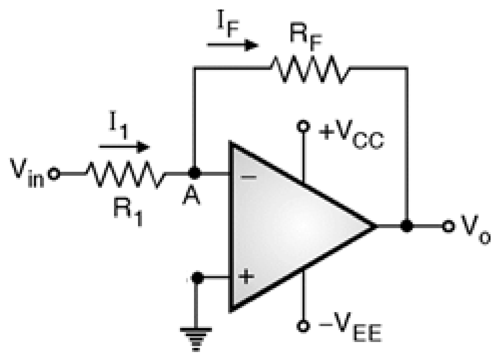
Comparison between positive and negative feedback :

Feature	**Negative Feedback**	**Positive Feedback**
Feedback Terminal	Inverting (–)	Non-inverting (+)
Effect on Gain	Reduces Gain	Increases Gain (to saturation)
Stability	Highly Stable	Unstable (Bistable)
Phase Shift	180 degree Out of phase	In phase
Primary Use	Linear Amplifiers	Oscillators, Schmitt Triggers
Linearity	Improves Linearity	Highly Non-linear

Applications of Op amp

Inverting Amplifier

- An op-amp connected as an Inverting Amplifier is shown in figure.
- Here input is applied to inverting terminal through a series input resistance R_1.
- And output is fed back through R_F to the same input.
- The non-inverting terminal is grounded.



Inverting Amplifier

- An expression for the output voltage of the inverting amplifier is :
- Apply KCL at node 'A'

$$I_1 = I_F$$

$$\frac{V_{in} - V_A}{R_1} = \frac{V_A - V_O}{R_F}$$

But, "A" is virtual ground point and $V_A = 0$…

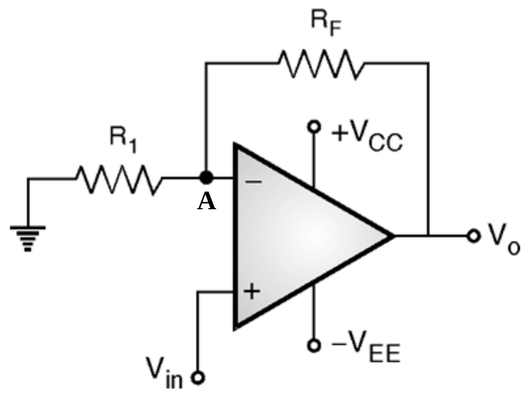
$$\boxed{V_0 = -\left(\frac{R_F}{R_1}\right) V_{in}}$$

The gain of circuit is decided by

$$\text{Voltage gain} = A_f = \frac{V_0}{V_{in}} = -\left(\frac{R_F}{R_1}\right)$$

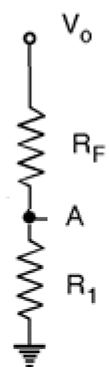
Non-Inverting Amplifier

- An op-amp connected as non-inverting amplifier is shown in figure.
- Here input is applied to non-inverting terminal.
- And output is applied back to the inverting terminal input through the feedback circuit formed by the input resistor R_1 and the feedback resistor R_F.
- This creates negative feedback.



Non-inverting Amplifier

- An expression for the output voltage of the non-inverting amplifier is :
- From figure,



- Here resistors R_1 and R_F form a voltage – divider circuit.

$$\therefore V_A = V_o\left(\frac{R_1}{R_1 + R_F}\right)$$

- But

$$V_A = V_{in}$$

$$\therefore V_{in} = V_o\left(\frac{R_1}{R_1 + R_F}\right)$$

- Rearranging the terms

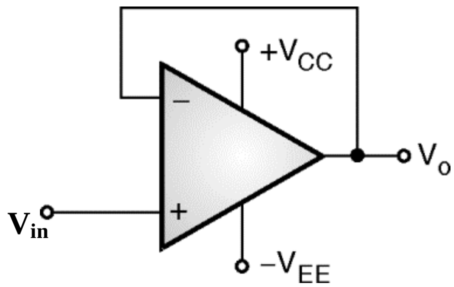
$$V_o = \left(\frac{R_1 + R_F}{R_1}\right) V_{in}$$

$$\boxed{\therefore V_o = \left(1 + \frac{R_F}{R_1}\right) V_{in}}$$

The gain of circuit is decided by

$$\text{Voltage gain} = A_f = \frac{V_o}{V_{in}} = \left(1 + \frac{R_F}{R_1}\right)$$

Buffer / Unity Gain Amplifier

- It is the special case of non- inverting amplifier.
- Here to achieve unity gain, we have to make $\mathbf{R_F = 0}$ (i.e. short circuit R_F) and $\mathbf{R_1} = \infty$ (i.e. open circuit R_1).
- Op-amp connected as non-inverting Amplifier with unity voltage gain is shown in figure.
- 

Buffer / Unity Gain Amplifier

- Here input is applied to non-inverting terminal.
- The output is same as applied input.
- This circuit is basically used for impedance matching purpose.
- It is also used for isolation purpose.
- It provides high input impedance and low output impedance hence avoids loading effect.
- An expression for the output voltage of the non-inverting amplifier is :
- We know for non – inverting amplifier V_o is expressed as

$$\therefore V_o = \left(1 + \frac{R_F}{R_1}\right) V_{in}$$

But, $R_F = 0$ and $R_1 = \infty$

$$\boxed{\therefore V_o = V_{in}}$$

OP-AMP Parameters

1) Input Offset Voltage

- The voltage which is applied between two input terminals of an op-amp so as to get zero output voltage is known as Input Offset Voltage.
- It is denoted by V_{io}.
- The smaller the value of V_{io}, the better the input terminals are matched.

Ideal Value	Practical Value For IC 741
0	2 mV

2) Output Offset Voltage

- It is the voltage appearing at the output of op-amp whenever both the input terminals are connected to ground.
- It is denoted by V_{OO}.
- The smaller the value of V_{oo}, the better performance of op-amp.
- This V_{OO} will be nullified using external compensation circuit consisting of variable resistor.

Ideal Value	Practical Value For IC 741
0	20 mV

3) Input Offset Current

- The algebraic difference between the currents entering into the inverting and non-inverting input terminal is referred to as Input Offset Current.
- It is denoted by I_{io} and expressed as

$$I_{io} = | I_{B1} - I_{B2} |$$

- Where I_{B1} is the current into the non-inverting input and I_{B2} is the current into the inverting input.
- The smaller the value of I_{io}, the better performance of op-amp.

Ideal Value	Practical Value For IC 741
0	6 ηA

4) Input Bias Current

- Input bias current is the average of the currents flowing into the inverting and non-inverting input terminal of an op-amp.
- It is denoted by I_B and expressed as

$$I_B = \frac{(I_{B1} + I_{B2})}{2}$$

- Where I_{B1} is the current into the non-inverting input and I_{B2} is the current into the inverting input.

- The smaller the value of I_B, the better performance of op-amp.

Ideal Value	Practical Value For IC 741
0	50 ηA

5) Input Impedance

- It is the resistance observed at any of the input terminal of an op-amp when another terminal is connected to ground.
- It is denoted by R_i.
- The input resistance should be as high as possible.

Ideal Value	Practical Value For IC 741
∞	2 MΩ

6) Output Impedance

- It is the equivalent resistance that can be measured between the output terminal of an op-amp and ground.
- It is denoted by R_O.
- The input resistance should be as small as possible.

Ideal Value	Practical Value For IC 741
0	75 Ω

7) Voltage Gain

- It is the ratio of amplified output of an op-amp to the differential input voltage applied to inputs of an op-amp.
- Since the amplitude of output signal is much larger than input signal, the voltage gain is commonly referred as LARGE SIGNAL VOLTAGE GAIN.
- It is denoted by A_v and expressed as
- Voltage Gain = $A\nu = \dfrac{\text{Output Voltage } (V_o)}{\text{Differential Input Voltage } (V_{id})}$
- The voltage gain should be as high as possible.

Ideal Value	Practical Value For IC 741
∞	2×10^5

8) Common Mode Rejection Ration (CMRR)

- It is defined as the ratio of the differential voltage gain (A_d) to the common mode gain (A_{cm})

- CMRR is given as

$$CMRR = \frac{A_d}{A_{cm}}$$

- It is the ability of an op-amp to reject common mode signals such as noise applied at both the input terminals.
- Ideally A_{cm} is zero and hence CMRR is ∞.
- CMRR is most often expressed in decibels (dB).
- The CMRR should be as high as possible.

Ideal Value	Practical Value For IC 741
∞	90 dB

9) Power Supply Rejection Ration (PSRR)

- It is also known as supply voltage rejection ration (SVRR).
- It is defined as the change in input offset voltage of an op-amp (ΔV_{io}) due to change in supply voltage (ΔV)
- CMRR is expressed either in microvolt per volt or decibels.
- PSRR is defined as ..

$$PSRR = \frac{\Delta V_{io}}{\Delta V}$$

- The lower the value of PSRR, the better for op-amp performance.

Ideal Value	Practical Value For IC 741
0	150 μV / V

10) Slew Rate

- It is the maximum rate of change of output voltage per unit time.
- Slew rate is expressed in volts per microseconds.
- SR is defined as ..

$$SR = \left.\frac{dV_o}{dt}\right|_{\text{maximum}}$$

- Higher the value of SR, better the performance of an op-amp.
- Thus slew rate decides the maximum frequency amplified by an op-amp without any distortion.

Ideal Value	Practical Value For IC 741
∞	0.5 V / μs

11) Gain-Bandwidth Product

- It is the bandwidth of an op-amp when the voltage gain is 1.
- It is also known as closed-loop bandwidth, unity gain bandwidth, and small-signal bandwidth.

Ideal Value	Practical Value For IC 741
∞	Approximately 1 MHz

Comparison of Ideal and Practical Parameters of an Op-amp

Sr. No.	Parameter	Ideal Value	Practical Value For IC 741
1.	Input Offset Voltage (V_{io})	0	2 mV
2.	Output Offset Voltage (V_{OO})	0	20 mV
3.	Input Offset Current (I_{io})	0	6 ηA
4.	Input Bias Current (I_B)	0	50 ηA
5.	Input Impedance (R_i)	∞	2 MΩ
6.	Output Impedance (R_O)	0	75 Ω
7.	Voltage Gain (A_v)	∞	2×10^5
8.	CMRR	∞	90 dB
9.	PSRR	0	150 µV / V
10.	Slew Rate	∞	0.5 V / µs
11.	Gain Bandwidth Product	∞	Approximately 1 MHz

Problem 1 : For an inverting amplifier, if R_F = 100KΩ, R_1 = 10KΩ, V_{CC} = ± 10V and Vin = 2V. Calculate output voltage V_O. Is this practically possible ? Justify your answer.

Solution :

Given,

$R_F = 100K\Omega$

$R_1 = 10K\Omega$

$V_{CC} = \pm 10V$

Vin = 2V

The output equation for inverting amplifier is ..

$$V_0 = -\left(\frac{R_F}{R_1}\right)V_{in}$$

$$V_O = -\left(\frac{100K\Omega}{10K\Omega}\right)2$$

$$\boxed{V_O = -20V}$$

V_O = -20V is practically not possible because the maximum output voltage we get is saturation voltage and is 90% of V_{CC} = 9V

Problem 2 : For non-inverting amplifier, if R_F = 20KΩ, R_1 = 1KΩ, V_{CC} = ± 12V and Vin = 3V. Calculate output voltage V_O. Comment on the result.

Solution :

Given,

$R_F = 20K\Omega$

$R_1 = 1K\Omega$

$V_{CC} = \pm 12V$

Vin = 3V

The output equation for inverting amplifier is ..

$$V_o = \left(1 + \frac{R_F}{R_1}\right)V_{in}$$

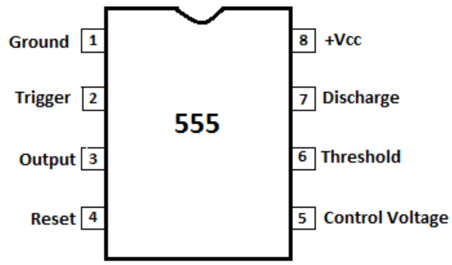
$$V_O = \left(1 + \frac{20K\Omega}{1K\Omega}\right)3$$

$$\boxed{V_O = 63\ V}$$

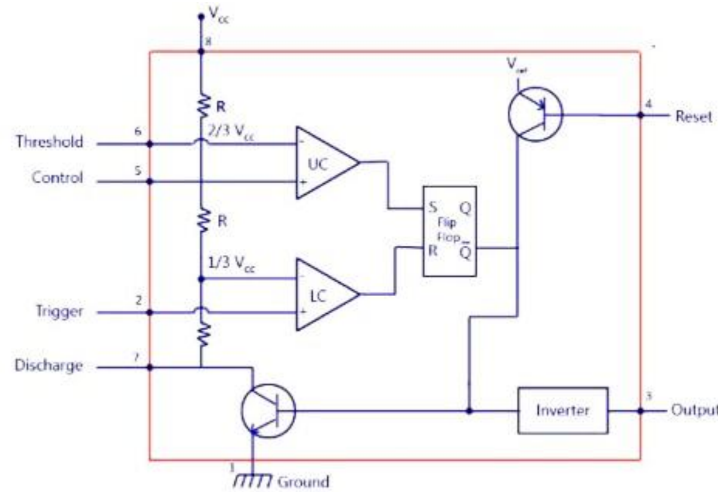
V_O = 63V is practically not possible because the maximum output voltage we get is saturation voltage and is 90% of V_{CC} = 10.8V

IC 555 :

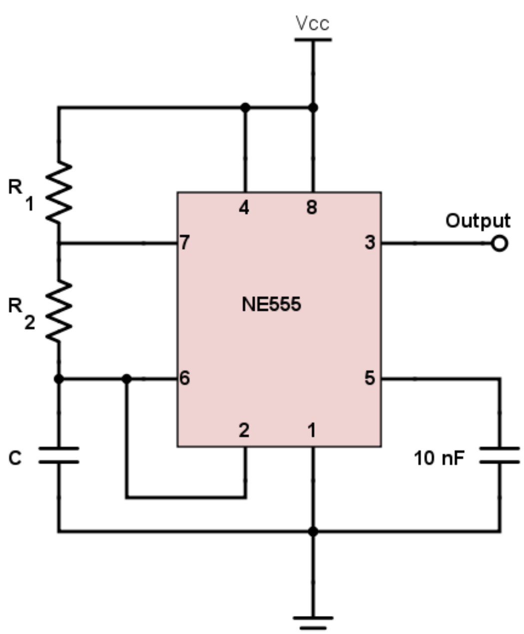
Pin diagram

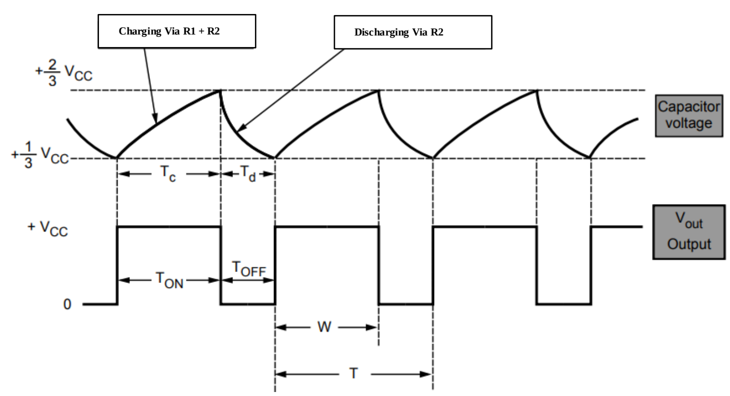


Internal block diagram



IC 555 as an Oscillator





- The Fig. shows the IC 555 connected as an Astable Multivibrator.
- The threshold input is connected to the trigger input.
- Two external resistances R_1, R_2 and a capacitor C is used in the circuit.
- This circuit has no stable state and hence called as Astable Multivibrator
- The circuits changes its state alternately, hence the operation is also called free running oscillator.

Working :

- The capacitor starts charging through the resistances R1, R2 and Vcc.
- As total resistance in the charging path is (R1 + R2), the charging time constant is (R1 + R2) C.
- The capacitor voltage is also given to threshold pin.
- While charging, capacitor voltage increases i.e. the threshold voltage increases.
- When it exceeds 2/3 Vcc, then the upper comparator output goes high which sets the flip-flop.
- The flip-flop output Q becomes high and $\bar{Q}$ becomes low.
- High Q drives discharge transistor Q and capacitor starts discharging through resistance R2 and discharge transistor Q.
- Thus the discharging time constant is R2 C.
- When capacitor voltage becomes less than 1/3 Vcc, lower comparator output goes high, resetting the flip-flop.
- Thus the flip-flop output Q goes low and $\bar{Q}$ goes high.
- The low Q makes the transistor off and capacitor starts charging
- This cycle repeats.
- Thus when capacitor is charging, output is high while when it is discharging the output is low.
- The output is a rectangular wave.

- The charging time for the capacitor is given by,
  T_{ON} = Charging time - 0.693 (R1 + R2) C
- While the discharge time is given by,
- T_{OFF} = Discharging time = 0.693 R2 C

RG 455

* Voltage Regulators :⇒

↳ Provides const. o/p vtg irrespective to change in Vin or I_L.

↳ Types of IC vtg. regulators.

linear Regu^r
① ↳ fixed o/p vtg. Regu : +/-ve o/p [78xx / 79xx]
② ↳ Adj. —〃— : +/-ve o/p [LM 317 / LM337]
③ ↳ Switching Regu. :

↳ ① & ② are linear Regu. The impedance of a linear regu.'s active element may be continuously varied to supply a desired ct. to load.

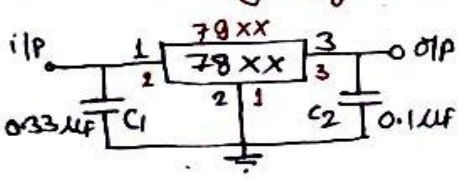
↳ ③ is switching regu^r ~~also~~ in which a sw. is turned ON & OFF at a rate such that the regu^r delivers the desired avg. ct. in periodic ~~pulsed~~ pulses to the load.

In case of switching regu^rs, the power discipation is very less ∴ they are more efficient than linear regu^r

But as most of the load cannot accept the avg. ct. in periodic pulses, most practical regu. are linear type.

↳ Applications :- ① & ② : in labs, for on card regulation.
③ : as control ckts in PWM.

* fixed vtg Regu :⇒ 1) +ve vtg Regu^r ② -ve vtg Regu.



↳ Vin(max) = 35V.

↳ I_o (max) = 1A

↳ 7805, 06, 08, 12, 15, 18. [7902, 5, 5.2, 6, 8, 12, 15, 18]
↳ Extra in 79xx.

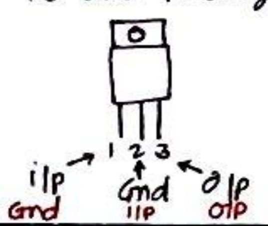
↳ Requires Heat sink.

↳ Vin − V_0 = Drop out vtg ≅ 2V (typically)

↳ C_1 is reqd. if regulator is located an appreciable distance from power supply filter.

↳ C_2 is used to ↑ve transient respo. of regu^r

TO-220 Package



• 78xx
• 79xx

* Performance Parameters of fixed vtg regu^r :⇒

① Line or I/P Regulation :⇒ change in V_o for change in V_i. Expressed in mV or as a % of V_o.

② Load Regulation :⇒ change in V_o for change in I_L. —"— same.

③ Temp. stability / Avg. temp. coefficient of V_o :⇒ change in V_o per unit temp. Expressed in either mV/°c or ppm/°c (Parts per million).

④ Ripple Rejection :⇒ It is the measure of a regulator's ability to reject ripple vtg. Expressed in dB.

* The smaller values of ①.②.③. the better regulator.

* Adjustable Vtg. Regu^r :⇒ LM317 ⇒ +ve ; LM337 ⇒ -ve.

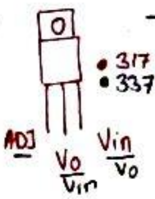
$V_O = 1.25\left(1+\frac{R_2}{R_1}\right)$

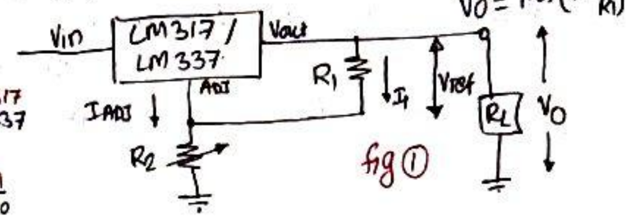
↳ V_o is 1.2V to 37V.

↳ $I_{O(max)}$ is 1.5 A.

↳ $V_{in(max)}$ = 40V

↳ Requires Heat Sink.





Adv. over fixed Regu:
↳ Improved system performance by having line + load regu^n of factor 10 or less.
↳ —"— overload protection
↳ —"— system reliability

↳ from fig ①
↳ Requires only 2 resi. to set V_o.
↳ Develops a nominal 1.25V, referred as V_{ref} [if $R_2 = 0$].
↳ V_{ref} is bet^n o/p & Adj terminals.
↳ Drop out vtg = $V_{in} - V_o \geq 3V$ But ≤ 40